

Bit-fields

Control and status registers pack flags and counters into one word

Extract, insert, mask

- Field width is hi minus lo plus one
- Bit zero is the LSB in this lab
- Before you OR an insert
- An unmasked insert value can spill into other fields
- Show the mask formula when you want the bit picture

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Bit-field extract / insert

Slice a word with [hi:lo], build masks, and **pack fields** into a register — the CSR / instruction habit.

Starter example: 16-bit CSR 0xA52D — ENABLE=1, MODE=2, COUNT=5, TAG=0xA5. Extract [7:3], then insert a new COUNT.

Load starter example

Challenges 0 / 22 cleared

Quiz: extract: Extract formula ends with? Answer: >> 1o

Answer

Show hintCheckNextIdle

Quiz: extractQuiz: widthQuiz: insertQuiz: bit 0Starter wordExtract COUNTExtract TAGExtract ENABLEInsert COUNTClear fieldShow maskPack CSRQuiz: MODE widthExtract MODESet wordQuiz: spillToggle bitQuiz: HDLInsert TAGRound-trip COUNTQuiz: packFull fields

Core ideas

Extract

(word & mask) >> 1o — isolate [hi:lo].

Insert

Clear the field with ~mask, then OR the shifted value.

Bit-fields starter

Workbook practice

- In the workbook track, take word A-five-two-D
- Extract COUNT at seven to three by hand and confirm five
- Clear that field with insert zero and check TAG is unchanged
- Write width of MODE at two to one as two
- Name one pitfall: inserting without masking so bits spill

Pitfalls to watch

- Do not assume bit zero is MSB, this lab uses LSB numbering
- Overlapping fields or wrong hi and lo break packs
- And remember: the browser lab is literacy
- Real CSRs still need a spec for reset values, access types, and side effects

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, extract and insert one field by hand
- When you are ready, take the short quiz, then continue to endian packing